

Smart Parking Management System

Project Proposal & Documentation

1. Problem Description

In large establishments, parking management is often unorganized and unpredictable. Drivers frequently face the frustration of entering a facility only to find no vacant spots, wasting time and fuel. Furthermore, managing different pricing tiers manually leads to errors and slow processing times.

This project solves these issues by providing a centralized dashboard to visualize slot availability in real-time and automate the registration and checkout process.

2. Functional Requirements

- **Real-time Slot Mapping:** A visual grid (A1 through F2) that changes color (Gray to Red) when occupied.
- **Multi-Tier User Management:**
 - *Regular Customers:* Standard fee of Php 50.00.
 - *Senior Citizens:* Discounted fee of Php 30.00.
- **Automated Data Handling:** Integrated DAO (Data Access Object) that routes information to specialized tables based on driver classification.
- **Transaction History:** Automatic archiving of records into a `transaction_history` table upon checkout to track total revenue.
- **Search & Filter:** Ability to locate drivers by Plate Number, Name, or Slot ID via the dashboard.

3. Key Features Implemented

Based on the system architecture, the project utilizes an **MVC-inspired design**:

- **GUI (Swing):** Interactive dashboard with real-time updates and popup dialogs for registration.
- **Database Integration:** Robust MySQL connectivity via `DBConnection` class.
- **Relationship Logic:** 1:N associations between the Dashboard and Reservations, ensuring data integrity across multiple tables.

4. Future Improvements

- **QR Code Integration:** Implementation of a simulated QR scan at the entrance to automate driver registration without manual data entry.
- **Mobile Companion App:** Allowing users to check slot availability before arriving at the establishment.
- **Automated SMS Notifications:** Notifying users of their parking duration or when their reserved time is nearing an end.
- **IoT Sensor Integration:** Using hardware sensors to detect vehicle presence automatically instead of manual button clicks.